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A Robust Patch-Based Hierarchical Framework for Image Forgery Detection and Classification

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1 Problem Definition & Originality

Problem Statement: With the rapid growth of deepfakes, AI-generated content, and digital misinformation, reliable image forgery detection has become critically important. Existing CNN-based and Transformer-based methods either focus only on spatial domain features or treat frequency/noise artifacts as simple preprocessing steps. This limits their ability to capture subtle forensic inconsistencies introduced during tampering.

The objective of this project is to design a robust, patch-based hierarchical framework that:

- Detects whether an image is authentic or manipulated.
- Classifies the type of forgery (Copy-Move, Splicing, Inpainting).
- Learns jointly from spatial, frequency, and noise forensic cues.

Originality Claim (Core Contribution):

We propose a **Three-Stream Tokenization Vision Transformer** where:

- Spatial patch embeddings,
- Frequency-domain (FFT magnitude) embeddings,
- Noise-residual (high-pass filtered) embeddings

are treated as **first-class tokens** and fused using a cross-attention mechanism before Transformer encoding.

Unlike existing works, frequency and noise are not used as preprocessing but are integrated directly into the Transformer token space, enabling unified multi-domain reasoning.

2 Literature Survey & Research Gap

Survey Scope: Analyzed traditional, CNN-based, and Transformer-based image forgery detection methods spanning block/keypoint, global CNN, patch-based CNN, standard ViTs, and frequency/noise forensic approaches.

Key Findings:

- **Block/Keypoint methods:** Sensitive to geometric transforms; fail in smooth regions.
- **Global CNN models:** Capture semantic features but miss subtle local forgery artifacts.
- **Patch-based CNNs:** Better local analysis, but limited to spatial domain only.
- **Standard ViTs:** Use single-stream spatial tokens — *ignore* frequency and noise forensic cues.
- **Frequency/Noise methods:** Effective individually, but never fused as *first-class tokens* inside a Transformer.

Limitations Identified:

- Sensitivity to post-processing (JPEG compression, blur, scaling).
- Lack of unified reasoning across spatial and forensic domains.
- Frequency/noise features used only as preprocessing.
- Standard ViTs rely purely on spatial tokens.

2.1 Comparison with Existing Work

Paper / Method	Primary Task	Supervision Type	Major Limitation
Block-based Methods (Fridrich, 2003)	Copy-Move Detection	Image-level / block matching	Sensitive to rotation, scaling, and compression
Frequency-aware CNN (Bayar & Stamm, 2016)	Forgery Detection	Image-level labels	Frequency used only as preprocessing, no joint learning
Standard Vision Transformer (Dosovitskiy et al., 2021)	Image Classification / Detection	Image-level labels	Uses only spatial tokens; ignores forensic noise and frequency cues
ManTra-Net (Wu et al., 2019)	Manipulation Localization	Pixel-level supervision	Feature concatenation only; no unified multi-domain reasoning
MVSS-Net (Dong et al., 2022)	Detection + Localization	Pixel-level supervision	Multi-view multi-scale CNN; no frequency-domain tokens or Transformer reasoning
TruFor (Guillaro et al., CVPR 2023)	Detection + Localization	Pixel-level supervision	Two-stream (RGB + noise) Transformer; lacks frequency-domain stream and cross-attention token fusion
Proposed Three-Stream ViT (Ours)	Detection + Classification	Type Image-level labels (patch aggregation)	Addresses limitation: Joint spatial, frequency, and noise reasoning via cross-attention

Table 1: Comparison of existing image forgery detection methods and our proposed approach.

Research Gap: No existing method jointly models spatial, frequency, and noise residual information as parallel token streams within a Vision Transformer using cross-attention fusion. This motivates our Three-Stream Tokenization Transformer framework.

3 Proposed Methodology (Technical Knowledge)

3.1 Overall Architecture

The proposed framework follows a systematic pipeline as illustrated in Figure 1:

1. Input image is resized and divided into non-overlapping patches.
2. For each patch, three parallel representations are computed:
 - Spatial representation (raw pixels)
 - Frequency representation (FFT magnitude spectrum)
 - Noise residual (high-pass filtered patch)
3. Each representation is linearly projected into embedding space.
4. Cross-attention fusion combines the three token types.
5. Fused tokens are passed to a Vision Transformer encoder.
6. Two-stage hierarchical classification is performed:
 - Stage 1: Binary Forgery Detection (Authentic vs. Tampered)
 - Stage 2: Forgery-Type Classification (Copy-Move, Splicing, Inpainting)

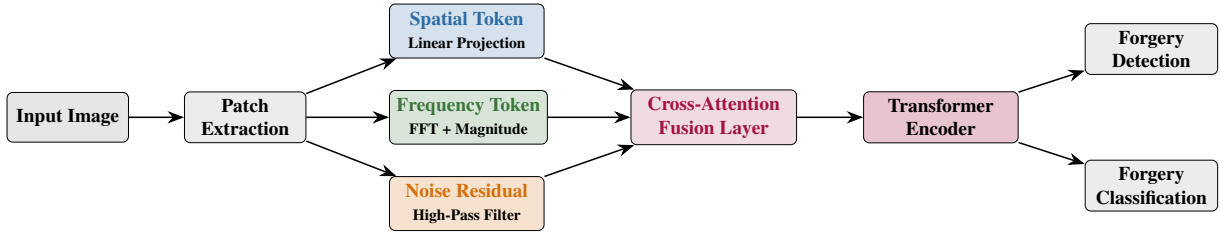


Figure 1: Proposed Three-Stream Tokenization Architecture. Each image patch generates three forensically-rich tokens (spatial, frequency, noise) that are fused via cross-attention before global Transformer reasoning.

3.2 Three-Stream Forensic Motivation

Each stream captures distinct and complementary forgery evidence:

- **Spatial Stream:** Captures raw pixel information, semantic content, visible artifacts, edge inconsistencies, and color/lighting mismatches at forgery boundaries.
- **Frequency Stream:** Extracts FFT magnitude spectrum revealing JPEG blocking artifacts, resampling traces, double compression signatures, and copy-move boundaries invisible in the RGB domain.
- **Noise Stream:** Computes high-pass noise residuals (SRM-style) to capture disrupted camera sensor noise patterns (PRNU) at forgery boundaries, following steganalysis-inspired filter design.

Key Insight: Forgery artifacts that are invisible in one domain are often prominent in another. Cross-attention fusion lets the Transformer attend to the most forensically relevant cues on a per-patch basis.

3.3 Mathematical Formulation

Let image patches be:

$$\{p_1, p_2, \dots, p_n\}$$

For each patch p_i , three token embeddings are computed:

$$z_s = W_s p_i \quad (\text{Spatial Token})$$

$$z_f = W_f |\mathcal{F}(p_i)| \quad (\text{Frequency Token})$$

$$z_n = W_n \text{HPF}(p_i) \quad (\text{Noise Residual Token})$$

Where:

- $\mathcal{F}$ denotes the 2D Fourier Transform
- $|\cdot|$ computes the magnitude spectrum
- HPF denotes the Gaussian High-Pass Filter (SRM-style)
- $W_s, W_f, W_n \in R^{d \times k}$ are learnable linear projection matrices

3.4 Cross-Attention Fusion

For each patch i , the three token embeddings are stacked:

$$\mathbf{T}_i = [z_s^{(i)}, z_f^{(i)}, z_n^{(i)}] \in R^{3 \times d}$$

Multi-head cross-attention is applied:

$$z_i^* = \text{CrossAttention}(\mathbf{T}_i)$$

Each stream attends to the other two streams, producing a single fused token $z_i^* \in R^d$ per patch.

Properties of the fusion layer:

- **Lightweight:** Single cross-attention layer (not a full Transformer).
- **Adaptive:** Learns which domain is most informative per patch.
- **Scalable:** Same d -dimensional output regardless of number of streams.
- **Differentiable:** End-to-end trainable with the main Transformer.

3.5 Hierarchical Classification

Fused tokens with positional embeddings are passed to the Transformer encoder, followed by:

- **Stage 1 — Binary Forgery Detection:** Classifies each patch (and aggregated image) as Authentic or Tampered.
- **Stage 2 — Forgery-Type Classification:** For detected tampered images, classifies the forgery type (Copy-Move, Splicing, Inpainting).

Final prediction with patch aggregation:

$$Y = A(f(z_1^*), f(z_2^*), \dots, f(z_n^*))$$

where A denotes aggregation via mean pooling or weighted voting, and f is the classification head.

3.6 Post-Processing Robustness (Data Augmentation)

To ensure the model learns forgery-intrinsic features rather than fragile compression or quality-dependent artifacts, the training dataset is augmented with realistic post-processing operations:

- JPEG recompression (QF 50–95)
- Gaussian blurring ($\sigma = 0.5$ – 2.0)
- Additive Gaussian noise ($\sigma = 1$ – 5)
- Random rescaling ($0.5\times$ – $1.5\times$)

This significantly improves cross-dataset generalization.

3.7 Explainability

- Transformer attention map visualization to identify forensically salient patches.
- Grad-CAM for patch importance mapping, enabling forensic interpretation of model decisions.

4 Datasets

The following datasets are selected for training and cross-dataset evaluation:

Dataset	Year	# Images	Forgery Types	Purpose
CASIA v2.0	2013	~12K	Copy-Move, Splicing	Primary training dataset
IMD2020	2020	~2010	Inpainting, Removal, Mixed	Training for inpainting and modern edits
Columbia	2004	363	Splicing	Cross-dataset evaluation (high-quality images)
COVERAGE	2018	200	Copy-Move	Cross-dataset evaluation (realistic scenarios)

Table 2: Datasets used for training and evaluation of the proposed framework.

5 Current Status & Future Plan

Current Status:

- Literature survey completed across traditional, CNN-based, and Transformer-based forgery detection methods.
- Research gap clearly identified — no prior work fuses spatial, frequency, and noise as parallel token streams inside a Transformer.
- Three-stream tokenization architecture designed with cross-attention fusion.
- Mathematical formulation finalized for all three streams and the fusion mechanism.

Next Steps (Phase 2):

- Dataset selection and preparation (CASIA v2.0, Columbia, COVERAGE, etc.).
- Implementation of three-stream tokenization pipeline.
- Cross-attention fusion module development.
- Model training with post-processing augmentation.
- Evaluation on in-dataset and cross-dataset benchmarks.
- Performance comparison with TruFor, MVSS-Net, and ManTra-Net.
- Ablation study on individual stream contributions.

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